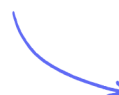


Proportion

Direct Proportion / Inverse Proportion / Variation Models

Medium (4 questions)	/12
Hard (4 questions)	/10
Very Hard (4 questions)	/18
Total Marks	/40

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Medium Questions

1 The table below shows the speed of a motorbike, v m/s at time t seconds.

t	0	1	2	3
v	0	2	8	18

(i) State which out of a linear or quadratic variation model would be most suitable for the data.

[1]

(ii) Find an equation for v in terms of t .

[2]

(3 marks)

2 y is inversely proportional to x^2 .
When $x = 4$, $y = 7.5$.
Find y when $x = 5$.

$y =$

(3 marks)

3 y varies inversely as x^2 .

When $x = 4$, $y = 2$.

Find y when $x = \frac{1}{2}$.

$y =$

(3 marks)

4 (a) A ball falls d metres in t seconds.

d is directly proportional to the square of t .

The ball falls 44.1 m in 3 seconds.

Find a formula for d in terms of t .

$d =$

(2 marks)

(b) Calculate the distance the ball falls in 2 seconds.

..... m

(1 mark)

Hard Questions

- 1 The table below shows the population of bacteria cells, p , in a petri dish x hours after their first introduction.

x	5	20	50
p	6	80	450

By finding and testing both a quadratic and a cubic variation model for p in terms of x , decide which would be a closer fit for the data.

(4 marks)

- 2 y is inversely proportional to the square root of x .
When $y = 7$, $x = 2.25$.
Write y in terms of x .

$y =$

(2 marks)

- 3 p is directly proportional to $(q + 2)^2$.
When $q = 1$, $p = 1$.

Find p when $q = 10$.

$p =$

(3 marks)

- 4 y varies directly with $(x - 1)^2$.
When $x = 5$, $y = 4$.
Find y when $x = 7$.

$y =$

(1 mark)

Very Hard Questions

- 1 (a) The intensity of a magnetic field, I , measured in Teslas, T, varies inversely as the cube of the distance from its source, d metres.

If the intensity of the magnetic field is 0.1 T at a distance of 2 metres, find an equation for the relationship between I and d .

(2 marks)

- (b) Complete the following table showing the intensity, I Teslas, at certain distances, d metres.

d	1	2	3	4
I		0.1		

(3 marks)

- (c) Use your model to calculate the distance from its source if $I = 0.0512$ metres.

(3 marks)

- 2 y is directly proportional to the cube root of $(x + 3)$.

When $x = 5$, $y = \frac{2}{3}$.

Find y when $x = 24$.

$y = \dots\dots\dots$

(3 marks)

- 3 (a)** t varies inversely with the square of $(x + 1)$.
When $x = 2$, $t = 5$.

Write t in terms of x .

$$t = \dots\dots\dots$$

(2 marks)

- (b)** When $t = 1.8$, find the positive value of x .

$$x = \dots\dots\dots$$

(2 marks)

- 4** y is inversely proportional to the square of $(x + 1)$.
 $y = 0.875$ when $x = 1$.

Find y when $x = 4$.

$$y = \dots\dots\dots$$

(3 marks)